gradbars: documented source

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1 Source organization and reproducibility

gradbars uses PGF/TikZ for inline data graphics. The editable runtime modules live in `sty/`; the generated `gradbars.dtx` bundles the same modules with this source guide. Generate it with `python scripts/build_source.py`. Check for drift with the `-check` option. The check extracts every module using docstrip and compares its contents with the runtime sources (line endings and trailing whitespace are normalized).

Run `tex gradbars.ins` in a separate directory containing the two distribution source files to extract all matching runtime modules. Install the complete set together. Run `pdflatex gradbars.dtx` from the repository root for this guide; the illustrated bilingual manuals are built separately by `python scripts/build_manual.py`. The default driver prints the guide only. Replace `\OnlyDescription` with `\AlsoImplementation` in a working copy to inspect the complete implementation with l3doc.

2 Four responsibilities

User commands and keys select data and scoped settings. The model validates input, resolves shared scales, and produces normalized geometry. The TikZ renderer draws paths and labels from that geometry. Integration modules manage alternative text, table measurement and external formatting.

The model uses normalized positions in $[0, 1]$ after the configured overflow policy. It retains raw input separately. Zero, start, stop and endpoint are not formatted labels. Projection to physical dimensions happens in the drawing font context so that `em`/`ex` retain their established meaning. The private `\__gradbars_render_tikz:` function is the default backend boundary, not a public callback. No second backend is supplied.

3 Stable extension contract

The public library, scalar type and numeric formatter registration commands documented in `docs/stability.md` are stable in version 1. Library registration is global and loading is confined to the preamble; repeated loading is harmless. Type and formatter names cannot be replaced. Scalar callbacks receive a validated number or `NA`. Formatter callbacks receive a display number, precision and number format; the engine appends units separately. Opaque formatters cannot participate in decimal alignment. Private variables and functions are not covered by the contract.

4 Validation and accessibility

Run `l3build check` and `python tests/run_tests.py` for semantic and three-engine regression tests. Run `python scripts/check_tagging.py` on a current tagging-capable TeX distribution to verify figure descriptions and measurement suppression. Automatic alternative text does not establish PDF/UA conformance for a complete table or document.

The optional `tblrlibgradbars.sty` is included in the same docstrip source. With tabularray 2025C or newer, `\UseTblrLibrary{gradbars}` loads cell helpers and dedicated columns. Its public trial hooks suspend tagging during measurement; fixed and X widths are tested in the PDF structure checks.

The Chinese and English manuals contain tutorials, option and library references, rendered examples and source. The developer reference documents the private data boundary; `docs/migration-v1.0.0.md` documents upgrade and compatibility rules. The build and archive checks are separate, allowing validation without creating a CTAN zip.